Rank of a Matrix

Echelon Form of a Matrix: A matrix $\boldsymbol{A}$ is said to in echelon form if

1. All the non-zero rows, if any precede the zero rows.
2. The number of zero preceding the first non-zero element in a row is less than the number of such zero in the succeeding row.

Example: $\begin{pmatrix} 0 & -6 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}$; $\begin{pmatrix} 5 & 4 & 0 & 1 \\ 0 & 0 & 2 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

Reduced Echelon Form of a Matrix: A matrix $\boldsymbol{A}$ is said to be in reduced echelon form if

1. All the non-zero rows, if any precede the zero rows.
2. The number of zero preceeding the first non-zero element in a row is less than the number of such zero in the succeeding row and
3. The first non-zero element in a row is unity.

Example: $\begin{pmatrix} 1 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

Row Reduced Echelon Form of a Matrix: A matrix $\boldsymbol{A}$ is said to be in row reduced echelon form if

1. $\boldsymbol{A}$ is an echelon form,
2. Each pivot element is 1 and
3. Each pivot element is the only non-zero entry in its columns.

Example: $\begin{pmatrix} 1 & 0 & 0 & 5 & 6 \\ 0 & 1 & 0 & 3 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

Pivot Element: The first non-zero entry of a non-zero row is called the pivot element of that row.

Example: $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{pmatrix}$; here, 1, 5 and 7 are the pivot elements of the 1^{st}, 2^{nd} and 3^{rd} rows respectively.

Normal Form of a Matrix: Every non-zero matrix ***A*** of order $m \times n$ can be reduced by the application of elementary row and column operations into equivalent matrix of one of the following forms:

$$\textbf{1.}\ \begin{pmatrix} I_r & O \\ O & O \end{pmatrix},\quad \textbf{2.}\begin{pmatrix} I_r \\ O \end{pmatrix},\quad \textbf{3.}\begin{pmatrix} I_r & O \end{pmatrix} \text{and} \textbf{4.}\begin{pmatrix} I_r \end{pmatrix}$$

where I_r is $r \times r$ identity matrix and O null matrix of any order.

These four forms are called normal or canonical form of ***A.***

Elementary Operations or Transformation on a Matrix: Let ***A*** be any matrix. Consider the following operations on the matrix ***A***.

1. Interchange of any two rows or two columns of ***A***.
2. Multiply each of the element of a row or a column of ***A*** with a non-zero scalar.
3. Add scalar multiple of one row of ***A*** to another row or one column of ***A*** two another column.

Each of the above operations are called elementary transformation or operations of ***A***.

Rank of a Matrix: Let ***M*** be a matrix. The number of pivot elements in an echelon form of ***M*** is called the rank of ***M*** and it is denoted by $\rho(M)$.

Alternatively, the number of non-zero rows in a row echelon form of a matrix is called the rank of the matrix.

Problem 01: Find rank of the matrix **A** = $\begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ by row reduction .

Solution

Step 1 Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix} \underset{\sim}{\begin{matrix} R_3 \Leftarrow R_3 - R_1 \\ R_4 \Leftarrow R_4 - R_1 \end{matrix}} \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 6 & 8 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 4 & 1 \\ 0 & 6 & 8 & 2 \end{bmatrix} \underset{\sim}{\begin{matrix} R_3 \Longleftarrow 3 \times R_3 - 4 \times R_2 \\ R_4 \Longleftarrow R_4 - 2 \times R_2 \end{matrix}} \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 0 & 0 & -4 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 2 **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 3.

Problem 02: Find rank of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{bmatrix}$ by row reduction .

Solution

Step 1 Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{bmatrix} \underset{\sim}{\begin{matrix} R_2 \Longleftarrow R_2 - 2 \times R_1 \\ R_3 \Longleftarrow R_3 - R_1 \\ R_4 \Longleftarrow R_4 - 2 \times R_1 \end{matrix}} \begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 5 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 5 & 1 \end{bmatrix} \underset{\sim}{R_2 \Longleftrightarrow R_4} \begin{bmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 2 **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 3.

Problem 03: Find rank of the matrix $\mathbf{A} = \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \end{bmatrix}$.

$$\begin{bmatrix} 0 & 1 & 2 & 1 \end{bmatrix}$$

Solution

Step 1 Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix} \underset{\sim}{R_3 \Leftarrow= 3 \times R_3 - R_1} \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & -4 & -9 & 7 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & -4 & -9 & 7 \\ 0 & 1 & 2 & 1 \end{bmatrix} \begin{matrix} R_3 \Leftarrow= R_3 + 2 \times R_2 \\ R_4 \Leftarrow= 2 \times R_4 - R_2 \\ \sim \end{matrix} \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 5 & -9 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 5 & -9 \\ 0 & 0 & 2 & 1 \end{bmatrix} \underset{\sim}{R_4 \Leftarrow= 5 \times R_4 - 2 \times R_3} \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 5 & -9 \\ 0 & 0 & 0 & 23 \end{bmatrix}$$

Step 2 **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 4.

Problem 04: Find rank of the matrix **A** = $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 6 & 7 & 4 \\ 4 & 7 & 10 & 13 & 16 \\ 5 & 8 & 9 & 10 & 3 \end{bmatrix}$.

Solution

Step 1 Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{matrix} R_2 \Leftarrow= R_2 - 2 \times R_1 \\ R_3 \Leftarrow= R_3 - 3 \times R_1 \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 6 & 7 & 4 \\ 4 & 7 & 10 & 13 & 16 \\ 5 & 8 & 9 & 10 & 3 \end{bmatrix} \begin{matrix} R_4 \Leftarrow= R_4 - 4 \times R_1 \\ R_5 \Leftarrow= R_5 - 5 \times R_1 \\ \sim \\ \ \end{matrix} \begin{bmatrix} 0 & -1 & -2 & -3 & -4 \\ 0 & -1 & -3 & -5 & -11 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -2 & -6 & -10 & -22 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -1 & -3 & -5 & -11 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & -2 & -6 & -10 & -22 \end{bmatrix} \begin{matrix} R_3 \Leftarrow= R_3 - R_2 \\ R_4 \Leftarrow= R_4 - R_2 \\ R_5 \Leftarrow= R_5 - 2 \times R_2 \\ \sim \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & 0 & 1 & 2 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 4 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & 0 & 1 & 2 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 4 & 14 \end{bmatrix} \begin{matrix} R_5 \Leftarrow= R_5 - 2 \times R_3 \\ \sim \end{matrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -1 & -2 & -3 & -4 \\ 0 & 0 & 1 & 2 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

<u>Step 2</u> **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 3.

<u>Problem 05:</u> Find rank of the matrix **A** = $\begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 1 & 3 & -2 & 0 & -4 \\ 3 & 8 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{bmatrix}$.

<u>Solution</u>

<u>Step 1</u> Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 1 & 3 & -2 & 0 & -4 \\ 3 & 8 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{bmatrix} \begin{matrix} R_2 \Leftarrow= R_2 - R_1 \\ R_3 \Leftarrow= R_3 - 3 \times R_1 \\ R_4 \Leftarrow= R_4 - 2 \times R_1 \\ \sim \end{matrix} \begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -3 & -3 & -6 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 2 & 2 & 4 & -2 \\ 0 & -3 & -3 & -6 & 3 \end{bmatrix} \begin{matrix} R_3 \Leftarrow\!= R_3 - 2 \times R_2 \\ R_4 \Leftarrow\!= R_4 + 3 \times R_2 \\ \sim \end{matrix} \begin{bmatrix} 1 & 2 & -3 & -2 & -3 \\ 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 2 **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 2.

Problem 06: Find rank of the matrix **A** = $\begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ by row reduction .

Solution

Step 1 Transform the matrix **A** to **row echelon** form.

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} R_1 \Leftarrow\!=\!\Rightarrow R_2 \\ \sim \end{matrix} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} R_4 \Leftarrow\!= R_4 - R_1 \\ \sim \end{matrix} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_3 \Leftarrow\!= R_3 - R_2 \\ \sim \end{matrix} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 2 **Rank** equals **number of nonzero rows** in the echelon form of the matrix.

Therefore, rank of **A** = 4.

Problem-07: Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrix. $A=\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 2 & 4 & 1 & -2 & 3\\ 3 & 6 & 2 & -6 & 5\end{pmatrix}$

Solution: Given that,

$$A=\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 2 & 4 & 1 & -2 & 3\\ 3 & 6 & 2 & -6 & 5\end{pmatrix}$$

Now we shall reduce the matrix to echelon form by the elementary row operations.

$$\therefore \quad A=\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 2 & 4 & 1 & -2 & 3\\ 3 & 6 & 2 & -6 & 5\end{pmatrix}$$

$$\approx\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 0 & 0 & 3 & -6 & 1\\ 0 & 0 & 5 & -12 & 2\end{pmatrix} \qquad \begin{matrix}R_2^{'}=R_2-2R_1\\ R_3^{'}=R_3-3R_1\end{matrix}$$

$$\approx\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 0 & 0 & 3 & -6 & 1\\ 0 & 0 & 0 & -6 & 1\end{pmatrix} \qquad R_3^{'}=3R_3-5R_2$$

This matrix is in echelon form.

Since it contains 3 non-zero rows so the rank of the matrix ***A*** is, $\rho(A)=3.$

Again,

$$\approx\begin{pmatrix}1 & 2 & -1 & 2 & 1\\ 0 & 0 & 1 & -2 & \frac{1}{3}\\ 0 & 0 & 0 & 1 & -\frac{1}{6}\end{pmatrix} \qquad \begin{matrix}R_2^{'}=\frac{1}{3}R_2\\ R_3^{'}=-\frac{1}{6}R_3\end{matrix}$$

$$\approx \begin{pmatrix} 1 & 2 & 0 & 0 & \frac{4}{3} \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 0 & 1 & -\frac{1}{6} \end{pmatrix} \quad R_1' = R_1 + R_2$$

$$\approx \begin{pmatrix} 1 & 2 & 0 & 0 & \frac{4}{3} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\frac{1}{6} \end{pmatrix} \quad R_2' = R_2 + 2R_3$$

This matrix is in row reduced echelon form.

<u>Problem-08:</u> Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrix. $A = \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$

Solution: Given that,

$$A = \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$$

Now we shall reduce the matrix to echelon form by the elementary row operations.

$$\therefore \quad A = \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$$

$$\approx \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 4 & 9 & -7 \\ 0 & 1 & 2 & 1 \end{pmatrix} \quad R_3' = R_1 - 3R_3$$

$$\approx \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 5 & -9 \\ 0 & 0 & -2 & -1 \end{pmatrix} \begin{matrix} R_3^{'} = R_3 - 2R_2 \\ R_4^{'} = R_2 - 2R_4 \end{matrix}$$

$$\approx \begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 5 & -9 \\ 0 & 0 & 0 & -23 \end{pmatrix} \quad R_4^{'} = 5R_4 + 2R_3$$

This matrix is in echelon form.

Since it contains 4 non-zero rows so the rank of the matrix ***A*** is, $\rho(A) = 4.$

Again,

$$\approx \begin{pmatrix} 1 & -\frac{2}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 1 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{9}{5} \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \begin{matrix} R_1^{'} = \frac{R_1}{3} \\ R_2^{'} = \frac{R_2}{2} \\ R_3^{'} = \frac{R_3}{5} \\ R_4^{'} = \frac{-R_4}{23} \end{matrix}$$

$$\approx \begin{pmatrix} 1 & 0 & \frac{2}{3} & 0 \\ 0 & 1 & 1 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{9}{5} \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad R_1^{'} = R_1 + \frac{2R_2}{3}$$

$$\approx \begin{pmatrix} 1 & 0 & 0 & \frac{6}{5} \\ 0 & 1 & 0 & \frac{23}{10} \\ 0 & 0 & 1 & -\frac{9}{5} \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \begin{matrix} R_1^{'} = R_1 - \frac{2R_3}{3} \\ R_2^{'} = R_2 - R_3 \end{matrix}$$

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \begin{matrix} R_1^{'} = R_1 - \frac{6R_4}{5} \\ R_2^{'} = R_2 - \frac{23R_4}{10} \\ R_3^{'} = R_3 + \frac{9R_4}{5} \end{matrix}$$

This is row reduced echelon form of the matrix ***A***.

This is also normal form of the matrix ***A.***

$$i.e, \left(I_4\right)$$

Problem-09: Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrix.

$$A = \begin{pmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}$$

Solution: Given that,

$$A = \begin{pmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}$$

Now we shall reduce the matrix to echelon form by the elementary row operations.

$$\therefore \ A = \begin{pmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}$$

$$\approx \begin{pmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 5 & 1 \end{pmatrix} \quad \begin{matrix} R_2^{'} = R_2 - 2R_1 \\ R_3^{'} = R_3 - R_1 \\ R_4^{'} = R_4 - 2R_1 \end{matrix}$$

$$\approx \begin{pmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_2 \leftrightarrow R_4$$

This matrix is in echelon form.

Since it contains 3 non-zero rows so the rank of the matrix ***A*** is, $\rho\left(A\right) = 3.$

$$\approx \begin{pmatrix} 1 & 3 & -2 & -1 \\ 0 & 0 & 1 & \frac{1}{5} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_2' = \frac{1}{5} R_2 \\ R_3' = \frac{1}{2} R_3 \end{matrix}$$

$$\approx \begin{pmatrix} 1 & 3 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & \frac{1}{5} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_1' = R_1 + 2R_2$$

$$\approx \begin{pmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_1' = R_1 + \frac{3}{5} R_3 \\ R_2' = R_2 - \frac{1}{5} R_3 \end{matrix}$$

This is row reduced echelon form of the matrix ***A***.

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_2' = C_2 - 3C_1$$

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_2 \leftrightarrow C_3$$

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad C_3 \leftrightarrow C_4$$

$$\approx \begin{pmatrix} I_3 & O \\ O & O \end{pmatrix}$$

This is the normal form of the matrix ***A***.

Problem-10: Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrix.

$$A=\begin{pmatrix}1 & 3 & 1 & -2 & -3\\ 1 & 4 & 3 & -1 & -4\\ 2 & 3 & -4 & -7 & -3\\ 3 & 8 & 1 & -7 & -8\end{pmatrix}$$

Solution: Given that,

$$A=\begin{pmatrix}1 & 3 & 1 & -2 & -3\\ 1 & 4 & 3 & -1 & -4\\ 2 & 3 & -4 & -7 & -3\\ 3 & 8 & 1 & -7 & -8\end{pmatrix}$$

Now we shall reduce the matrix to echelon form by the elementary row operations.

$$\therefore\ A=\begin{pmatrix}1 & 3 & 1 & -2 & -3\\ 1 & 4 & 3 & -1 & -4\\ 2 & 3 & -4 & -7 & -3\\ 3 & 8 & 1 & -7 & -8\end{pmatrix}$$

$$\approx\begin{pmatrix}1 & 3 & 1 & -2 & -3\\ 0 & 1 & 2 & 1 & -1\\ 0 & -3 & -6 & -3 & 3\\ 0 & -1 & -2 & -1 & 1\end{pmatrix}\qquad \begin{matrix}R_2^{'}=R_2-R_1\\ R_3^{'}=R_3-2R_1\\ R_4^{'}=R_4-3R_1\end{matrix}$$

$$\approx\begin{pmatrix}1 & 3 & 1 & -2 & -3\\ 0 & 1 & 2 & 1 & -1\\ 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0\end{pmatrix}\qquad \begin{matrix}R_3^{'}=R_3+3R_2\\ R_4^{'}=R_4+R_2\end{matrix}$$

This matrix is in echelon form.

Since it contains 2 non-zero rows so the rank of the matrix $\boldsymbol{A}$ is, $\rho(A)=2.$

$$\approx \begin{pmatrix} 1 & 0 & -5 & -5 & 0 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad R_1^{'} = R_1 - 3R_2$$

This is row reduced echelon form of the matrix ***A***.

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad \begin{matrix} C_3^{'} = C_3 + 5C_1 \\ C_4^{'} = C_4 + 5C_1 \end{matrix}$$

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad \begin{matrix} C_3^{'} = C_3 - 2C_2 \\ C_4^{'} = C_4 - C_2 \\ C_5^{'} = C_5 + C_2 \end{matrix}$$

$$\approx \begin{pmatrix} I_2 & O \\ O & O \end{pmatrix}$$

This is the normal form of the matrix ***A***.

Problem-11: Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrix.

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 5 & 3 & 5 & 3 \end{pmatrix}$$

Solution: Given that,

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 5 & 3 & 5 & 3 \end{pmatrix}$$

Now we shall reduce the matrix to echelon form by the elementary row operations

$$\therefore \quad A=\begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 5 & 3 & 5 & 3 \end{pmatrix}$$

$$\approx \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & -4 & 0 & -4 \\ 0 & -3 & 0 & -3 \\ 0 & -7 & 0 & -7 \end{pmatrix} \qquad \begin{aligned} R_2' &= R_2 - 2R_1 \\ R_3' &= R_3 - 3R_1 \\ R_4' &= R_4 - 3R_1 \\ R_5' &= R_5 - 5R_1 \end{aligned}$$

$$\approx \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \begin{aligned} R_3' &= 3R_3 - 4R_2 \\ R_4' &= R_4 - R_2 \\ R_5' &= 3R_5 - 7R_2 \end{aligned}$$

This matrix is in echelon form.

Since it contains 2 non-zero rows so the rank of the matrix ***A*** is, $\rho(A) = 2$.

$$\approx \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad R_2' = -\frac{1}{3}R_2$$

$$\approx \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad R_1' = R_1 - 2R_2$$

$$\approx \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_1^{'} = R_1 - 2R_2$$

This is row reduced echelon form of the matrix $\boldsymbol{A}$.

$$\approx \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} C_3^{'} = C_3 - C_1 \\ C_4^{'} = C_4 - C_2 \end{matrix}$$

$$\approx \begin{pmatrix} I_2 & O \\ O & O \end{pmatrix}$$

This is the normal form of the matrix $\boldsymbol{A}$.

Exercise: 02

1. Find the Rank, Echelon form(EF), Row reduced echelon form(RREF) and Normal form(NF) of the following matrices:

i) $\begin{pmatrix} 1 & 2 & 3 & 2 & 1 \\ 3 & 1 & -5 & -2 & 1 \\ 7 & 8 & -1 & 2 & 5 \end{pmatrix}$, **ii)** $\begin{pmatrix} 3 & 2 & 1 & 3 & 0 \\ -1 & 2 & 4 & 6 & 7 \\ 0 & 2 & 2 & 4 & 4 \end{pmatrix}$ **iii)** $\begin{pmatrix} 4 & -2 & 8 & 0 & 4 \\ 1 & -1 & 9 & 0 & -5 \\ 0 & -7 & 5 & 0 & 2 \end{pmatrix}$

iv) $\begin{pmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{pmatrix}$ **v)** $\begin{pmatrix} 1 & -1 & 2 & 1 \\ 3 & 0 & 2 & 2 \\ 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$ **vi)** $\begin{pmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$

vii) $\begin{pmatrix} -2 & 1 & 3 & 4 \\ 5 & 0 & -1 & -1 \\ 3 & 1 & 2 & 3 \\ 1 & 0 & 1 & 3 \end{pmatrix}$ **viii)** $\begin{pmatrix} 1 & 2 & -3 & -2 & -3 \\ 1 & 3 & -2 & 0 & -4 \\ 3 & 8 & -7 & -2 & -11 \\ 2 & 1 & -9 & -10 & -3 \end{pmatrix}$ **ix)** $\begin{pmatrix} -1 & 3 & 1 \\ 7 & 5 & 6 \\ 2 & -2 & 0 \\ 4 & 2 & 3 \\ 3 & -9 & -3 \end{pmatrix}$